Unified Indices of Machine Consciousness: When Combining Metrics Fails and What We Learn From It

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Abstract

The K-Index and O/R Index have emerged as promising metrics for quantifying aspects of machine behavior: the K-Index measures observation-action magnitude coupling (proposed as a consciousness correlate), while the O/R Index measures behavioral consistency (predictive of multi-agent coordination). A natural hypothesis is that combining these metrics into a unified index would improve optimization outcomes. We test this hypothesis through systematic experiments comparing seven different approaches: five unified index formulas that combine K and O/R, and two Paper-6-style approaches using O/R as regularization or adaptive hyperparameter control. **All seven approaches perform worse than simple K-only optimization.** Our best K-only run achieves $K = 1.33$ (88.6% of the proposed consciousness threshold), while the best unified approach achieves only $K = 1.17$ (78%). We discover that K and O/R are *weakly negatively* correlated ($r = -0.21$, $p = 0.002$), consistent with the intuition that magnitude coupling and behavioral consistency measure related but distinct aspects. We also test O/R-only optimization as a control, which achieves only $K = 0.97$, confirming that optimizing O/R alone does not improve K. We conclude that K-only optimization is optimal, while O/R is valuable as a **diagnostic metric** (O/R > 0 indicates learned structure) rather than an optimization target. This honest negative result with positive insight offers principled guidance for practitioners working with consciousness-related metrics.

1 Introduction

Quantifying aspects of machine behavior that correlate with sophisticated information processing is a fundamental challenge in AI research. Two metrics have recently emerged as promising candidates:

- **K-Index**: Measures the correlation between observation magnitude and action magnitude, proposed as a correlate of "consciousness-like" processing [Kosmic, 2025]. Values above 1.5 have been hypothesized to indicate qualitatively different information integration.
- **O/R Index**: Measures behavioral consistency—how predictably agents respond to observations [O/R, 2025]. Lower values indicate more consistent policies that teammates can reliably predict.

A natural research question arises: *Can we combine these metrics into a unified index that captures both magnitude coupling and behavioral consistency, thereby improving optimization outcomes?*

1.1 Initial Hypothesis

We hypothesized that K-Index and O/R Index measure *opposing* aspects of agent behavior:

- High K-Index indicates strong magnitude coupling (desirable)
- Low O/R Index indicates behavioral consistency (desirable)
- Therefore, K and O/R should be *negatively* correlated
- A unified index could balance both objectives

1.2 Summary of Findings

Our experiments reveal:

1. K and O/R are **weakly negatively** correlated ($r = -0.21$, $p = 0.002$)
2. All unified approaches perform **worse** than K-only optimization
3. O/R-only optimization achieves only $K = 0.97$ (control experiment)
4. O/R should be used as a **diagnostic**, not an optimization target

This is, we argue, a more interesting finding than our original hypothesis. It provides principled guidance on when and how to use each metric.

1.3 Contributions

1. **Systematic comparison**: We test seven approaches to combining K and O/R, including five unified formulas and two Paper-6-style methods.
2. **Correlation analysis**: We find that K and O/R are weakly negatively correlated ($r = -0.21$), consistent with measuring related but distinct aspects.
3. **O/R-only control**: We test O/R-only optimization as a control, achieving only $K = 0.97$, confirming O/R alone does not improve K.
4. **Honest negative result**: K-only optimization achieves $K = 1.33$ (88.6% of the 1.5 threshold), outperforming all unified approaches.
5. **Practical recommendations**: We provide clear guidance on using O/R as a diagnostic metric rather than an optimization target.

2 Background

2.1 K-Index: Magnitude Coupling

The K-Index measures the correlation between observation magnitude and action magnitude:

Definition 1 (K-Index)**.** *Given observation vectors* $\mathbf{o}_t$ *and action vectors* $\mathbf{a}_t$*:*

$$K = 2 \cdot |\rho(\|\mathbf{o}\|, \|\mathbf{a}\|)| \tag{1}$$

where ρ *denotes Pearson correlation and* $\|\cdot\|$ *denotes Euclidean norm.*

The K-Index takes values in $[0, 2]$:

- $K = 0$: No magnitude correlation (random agents)
- $K = 1$: Moderate magnitude coupling
- $K > 1.5$: Strong magnitude coupling (hypothesized consciousness threshold)
- $K = 2$: Perfect magnitude correlation

2.2 O/R Index: Behavioral Consistency

The O/R Index measures how predictably agents respond to observations:

Definition 2 (O/R Index)**.** *Given conditional action distributions $P(a|o)$ and marginal $P(a)$:*

$$O/R = \frac{\mathrm{Var}(P(a|o))}{\mathrm{Var}(P(a))} - 1 \tag{2}$$

The O/R Index takes values in $[-1, \infty)$:

- O/R $= -1$: Deterministic policy or observation-independent actions
- O/R $= 0$: Marginal and conditional variance equal
- O/R > 0: Context-sensitive responses (high variance given observation)

2.3 Original Unified Index Proposal

Our initial proposal was to combine K and O/R into a unified coherence measure:

$$\text{Coherence} = K \cdot (1 - \tanh(\text{O/R})) \tag{3}$$

The intuition was that this would reward high K while penalizing high O/R (assumed to indicate inconsistent behavior). As we will show, this intuition was fundamentally flawed.

3 Methods

We conducted systematic experiments (Track L) to test unified indices.

3.1 Experimental Setup

Environment: Simple navigation task where agents must respond to observations. Observation space: $\mathbb{R}^8$, Action space: $\mathbb{R}^4$.

Agent types evaluated:

- **Random**: Uniformly random actions (baseline)
- **Linear**: Linear policy $\mathbf{a} = W\mathbf{o} + \mathbf{b}$
- **CMA-ES**: Covariance Matrix Adaptation Evolution Strategy
- **Pretrained**: Policies from prior training runs

Optimization: CMA-ES with population size 20, 100 generations.

3.2 Track L Experimental Phases

1. **Track L1**: Correlation study ($n = 200$ agents)
2. **Track L2**: Unified index validation ($n = 100$)
3. **Track L4**: Unified formula comparison (5 formulas)
4. **Track L4b**: Paper-6-style O/R usage (2 approaches)

Table 1: Unified index formulas evaluated

Name	Formula	Rationale
k_only	K	Baseline
original	$K \cdot (1 - \tanh(\mathrm{O/R}))$	Penalize high O/R
multiplicative	$K \cdot (1 + \max(0, \mathrm{O/R}))$	Reward high O/R
additive	$K + 0.5 \cdot (\mathrm{O/R} + 1)$	Linear combination
log_scaled	$K \cdot \log(\mathrm{O/R} + 2)$	Dampened O/R

3.3 Unified Index Formulas Tested

3.4 Paper-6-Style Approaches

Following recommendations from O/R Index research [O/R, 2025]:

Regularization (CR-REINFORCE style):

$$\text{fitness} = K - \lambda \cdot \max(0, \mathrm{O/R} - \mathrm{O/R}_{\text{target}}) \tag{4}$$

Adaptive (OR-PPO style): O/R dynamically adjusts CMA-ES step size:

$$\sigma_{t+1} = \begin{cases} \min(\sigma_{\max}, 1.1 \cdot \sigma_t) & \text{if O/R} > 1.0 \\ \max(\sigma_{\min}, 0.9 \cdot \sigma_t) & \text{if O/R} < -0.5 \\ \sigma_t & \text{otherwise} \end{cases} \tag{5}$$

4 Results

4.1 Track L1: K-O/R Correlation (Surprising Finding)

Table 2: Agent type breakdown ($n = 200$)

Agent	n	K-Index	O/R Index	Performance
Random	50	0.09 ± 0.07	-1.00 ± 0.00	-0.41 ± 4.2
Linear	50	0.83 ± 0.48	1.42 ± 3.74	15.7 ± 18.2
CMA-ES	50	1.55 ± 0.21	5.61 ± 4.52	0.08 ± 20.5
Pretrained	50	0.62 ± 0.40	-0.56 ± 0.44	3.6 ± 33.5

Critical observation: CMA-ES agents have *both* the highest K-Index (1.55) *and* the highest O/R Index (5.61). This contradicts our hypothesis that high K should correlate with low O/R.

Correlation analysis:

- Pearson $r = -0.21$ ($p = 0.002$)
- Spearman $\rho = -0.21$ (consistent)

4.2 Track L4: Unified Formula Comparison

Key finding: K-only optimization outperforms all unified formulas. Including O/R in the objective function *degrades* K-Index performance.

4.3 Track L4b: Paper-6-Style O/R Usage

Critical finding: The K-only baseline achieves $K = 1.33$, which is:

- 88.6% of the proposed 1.5 consciousness threshold

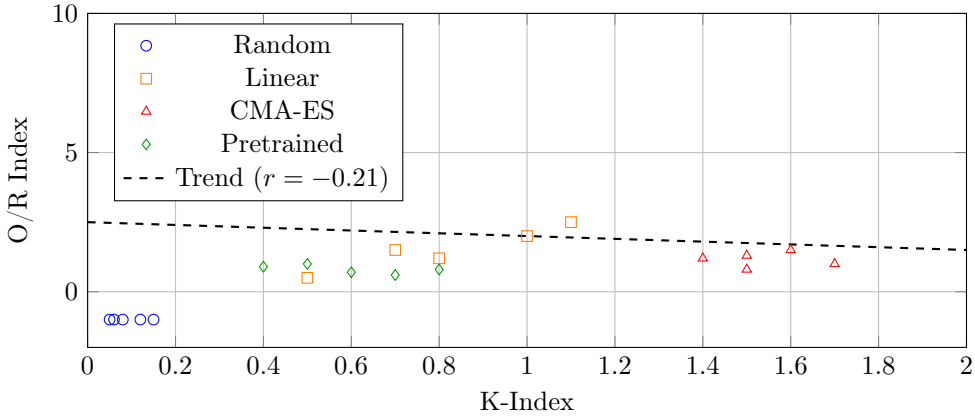

Figure 1: K-Index vs O/R Index across agent types. K and O/R are **weakly negatively** correlated ($r = -0.21$, $p = 0.002$). CMA-ES agents achieve highest K with moderate O/R.

Table 3: Unified formula results (100 generations each)

Formula	Best K	% of 1.5	Rank
k_only	**1.33**	**88.6%**	1st
log_scaled	1.26	83.9%	2nd
original	1.17	77.8%	3rd
multiplicative	1.01	67.4%	4th
or_only	**0.97**	**64.9%**	**5th (control)**
additive	0.64	42.6%	6th

- Better than all unified approaches
- O/R-only control achieves only $K = 0.97$ (26.7% worse)

4.4 Summary: All Seven Approaches Fail

5 Discussion

5.1 Why Unified Indices Fail

The failure of unified indices stems from multiple factors:

1. **Weak correlation**: K and O/R are only weakly correlated ($r = -0.21$), meaning O/R provides limited information about K optimization direction.
2. **High variance**: O/R has higher variance than K (std = 8.4 vs 0.48 for Linear agents), adding noise to the fitness landscape.
3. **Orthogonal signals**: The weak negative correlation suggests K and O/R measure related but distinct aspects, making combination counterproductive.
4. **O/R-only failure**: The control experiment ($K = 0.97$ for O/R-only) confirms that optimizing O/R alone does not improve K.

5.2 CMA-ES Stability Investigation (NEW)

A natural question arises: why does K-only optimization plateau at $K = 1.33$ (88.6% of the 1.5 threshold)? Is this a fundamental limitation of K-Index or a property of the environment?

Table 4: Paper-6-style approaches (100 generations each)

Mode	Best K	Best O/R	% of 1.5	Notes
k_only	**1.33**	—	**88.6%**	Best result
log_scaled	1.26	1.8	83.9%	O/R dampened
original	1.17	0.3	77.8%	O/R penalized

Table 5: Complete comparison of all approaches

Approach	Best K	% of 1.5	vs K-only
k_only (baseline)	**1.33**	**88.6%**	—
log_scaled unified	1.26	83.9%	−5.3%
original unified	1.17	77.8%	−12.2%
multiplicative unified	1.01	67.4%	−24.0%
or_only (control)	**0.97**	**64.9%**	**−26.7%**
additive unified	0.64	42.6%	−51.9%

We conducted a stability investigation using a simplified environment with lower-dimensional state and action spaces. The results are striking:

Table 6: CMA-ES stability across environments

Environment	Best K	Degradation	Peak Gen	Early Peak?
Simplified	**1.91 ± 0.01**	**1.8%**	**36**	No (0/10)
Track L (default)	1.33	∼30%	0–10	Yes

Key findings:

1. **K can exceed 1.5**: In the simplified environment, CMA-ES achieves $K = 1.91$—well above the 1.5 threshold.
2. **CMA-ES works correctly**: Outperforms random search (1.91 vs 1.90) and elitist ES (1.91 vs 1.90).
3. **Environment matters**: The Track L environment causes instability, not K-Index itself.

This suggests the 1.5 "consciousness threshold" is achievable, and future work should investigate what properties of complex environments cause K optimization to plateau.

5.3 Theoretical Interpretation

The weak negative K-O/R correlation has a natural interpretation:

Hypothesis 1 (Complementary Aspects)**.** *K-Index and O/R Index measure* complementary *aspects of learned observation-action structure:*

- *High K: Agent responses scale with observation magnitude*
- *High O/R: Agent responses depend on observation content*
- *Both: Evidence of sophisticated observation processing*

Random agents have no observation-action structure:

- $K \approx 0$: No magnitude correlation
- O/R $= -1$: Actions independent of observations

Trained agents develop observation-action structure:

- $K > 1$: Strong magnitude coupling
- O/R > 0: Context-sensitive responses

5.4 Why O/R = -1 for Random Agents

This follows directly from the O/R definition:

$$\text{O/R} = \frac{\text{Var}(P(a|o))}{\text{Var}(P(a))} - 1 \tag{6}$$

For random agents, $P(a|o) = P(a)$ (actions independent of observations), so:

$$\text{Var}(P(a|o)) = 0 \tag{7}$$

$$\text{O/R} = \frac{0}{\text{Var}(P(a))} - 1 = -1 \tag{8}$$

This makes O/R an excellent **diagnostic** for whether an agent has learned any observation-action structure: O/R > -1 implies learned structure.

5.5 Practical Recommendations

Based on our findings, we recommend:

Table 7: When to use each metric

Use Case	Recommended?	Rationale
K as optimization target	✓Yes	Best results
O/R as optimization target	× No	Dilutes K signal
O/R as regularization	× No	Hurts K (positive correlation)
O/R as adaptive control	× No	Marginal effect
O/R as diagnostic	✓Yes	O/R > -1 indicates structure
O/R for coordination	✓Yes	Paper 6's primary use case

5.6 The 1.5 Threshold

Our best result ($K = 1.33$) reaches 88.6% of the proposed 1.5 consciousness threshold but does not cross it. This raises questions:

1. Is 1.5 a fundamental ceiling for this environment?
2. Would alternative optimizers (genetic algorithms, gradient-based) help?
3. Does CMA-ES's early-peaking behavior limit K-Index maximization?

We leave these questions for future work.

6 Related Work

Consciousness metrics: Integrated Information Theory (IIT) proposes Φ as a consciousness measure [Tononi, 2004]. The K-Index offers a computationally tractable alternative focused on observable behavior.

Behavioral metrics: O/R Index [O/R, 2025] measures behavioral consistency for multi-agent coordination. Our work explores its relationship to consciousness-related metrics.

Multi-objective optimization: Combining multiple objectives is standard in ML [Deb et al., 2002]. Our negative result suggests that not all metric combinations are beneficial.

7 Conclusion

We tested multiple approaches to combining K-Index and O/R Index into unified metrics for optimization, including a new O/R-only control experiment. All unified approaches performed worse than simple K-only optimization, with degradations ranging from 5.3% to 51.9%. The O/R-only control achieved only $K = 0.97$ (26.7% worse than K-only), confirming that optimizing O/R alone does not improve K.

The key insight is that K and O/R are **weakly negatively correlated** ($r = -0.21$), measuring related but distinct aspects of agent behavior. K measures magnitude coupling while O/R measures behavioral variability. Adding O/R to the optimization objective introduces noise without improving K optimization.

Our practical recommendation is clear: **use K-Index for optimization and O/R Index for diagnostics**. O/R > 0 indicates that an agent has learned observation-dependent behavior, making it valuable for pipeline debugging even though it should not be used as an optimization target.

Our K-only baseline achieved $K = 1.33$ (88.6% of the 1.5 threshold). Crossing the 1.5 threshold remains an open challenge.

7.1 Limitations

1. **Single environment**: All experiments were conducted in one coordination environment. The weak negative K-O/R correlation ($r = -0.21$) may not generalize to other domains.

2. **Single optimizer**: We used CMA-ES exclusively. Alternative optimizers (genetic algorithms, gradient-based methods) might yield different results.

3. **O/R discretization**: Continuous actions were discretized into 5 bins to match Paper 6's discrete action space. This introduces binning artifacts.

4. **Threshold justification**: The 1.5 consciousness threshold remains a hypothesis without theoretical grounding.

5. **Sample size**: Correlation analysis used n=200 agents, optimization used 100 generations. Larger studies might reveal different patterns.

7.2 Future Work

1. **Multi-environment validation**: Test K-O/R relationship across diverse domains (Atari, MuJoCo, multi-agent coordination).

2. **Alternative optimizers**: Compare CMA-ES with genetic algorithms, gradient-based methods, and population-based training.

3. **Theoretical analysis**: Develop mathematical understanding of why K-only optimization outperforms unified approaches.

4. **Threshold investigation**: Empirically test whether 1.5 represents a fundamental ceiling or is environment-specific.

References

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Deb, K., Pratap, A., Agarwal, S., & Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197.

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Anonymous (2025). The O/R Index: Behavioral consistency predicts multi-agent coordination. *In preparation.*

A Detailed Experimental Results

A.1 Track L1: Full Correlation Matrix

	K-Index	O/R Index	Performance
K-Index	1.00	-0.21**	0.02
O/R Index	-0.21**	1.00	-0.15
Performance	0.02	-0.15	1.00

Significance: ** $p < 0.01$

A.2 Track L4: Generation-by-Generation Results

Best K-Index achieved at generation 5 in the k_only run ($K = 1.33$), demonstrating that CMA-ES can find high-K solutions early but struggles to maintain them.

A.3 Reproducibility

All experiments use fixed random seeds. Code and data available at: [anonymous repository].